

Diagonal Polynomials and Dyadic Block Geometry in Repeated Thue–Morse Prefix Summation

Exact formulas, denominator laws, rational roots, and fast Wolfram Language evaluation

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3 September 2026

Abstract

Let $\varepsilon_m = (-1)^{\text{wt}_2(m)}$ be the signed Thue–Morse sequence and let σ_r be its r -fold inclusive prefix sum. We analyze the Wolfram Language table in the prompt, whose recursive step convolves one row with the weights $1, 2, 3, \dots$. The first structural result is the exact identification

$$s_{n,k} = \sigma_{2n+1}(k - n - 1),$$

where negative arguments of σ_r are interpreted as zero. Thus one row step performs two further prefix summations and shifts the column by one. For a fixed diagonal offset $d \geq 1$, write $q = d - 1$. Then

$$s_{n,n+d} = D_q(n), \quad D_q(x) = \sum_{j=0}^{\lfloor q/2 \rfloor} \varepsilon_j \binom{2x + q - 2j - 1}{q - 2j}$$

with the binomial understood polynomially. Equivalently,

$$\sum_{q \geq 0} D_q(x) z^q = \frac{\mathcal{K}(z^2)}{(1 - z)^{2x}}, \quad D_q(x) = \sum_{j=0}^{\lfloor q/2 \rfloor} \varepsilon_j \frac{(2x)^{\overline{q-2j}}}{(q - 2j)!}.$$

This is the requested general shape: a sparse, Thue–Morse-signed expansion in rising factorials of degrees $q, q - 2, q - 4, \dots$. It immediately gives $\deg D_q = q$, leading coefficient $2^q/q!$, exact monomial coefficients through unsigned Stirling numbers, and fast evaluation in $O(q)$ exact-arithmetic terms.

We also prove several finer facts. The least positive integer clearing all monomial coefficients is

$$\frac{q!}{2^{\min(\nu_2(q!), \lceil q/2 \rceil)}}.$$

Every rational root of D_q is an integer or half-integer. All nonnegative rational roots are classified exactly: $x = m/2$, $m \geq 0$, is a root if and only if

$$q \bmod 2^{m+1} \geq 2^{m+1} - (m + 1).$$

Finally, a finite positive pulse polynomial describes every row. Its coefficients are positive, palindromic, unimodal, have an exactly flat central plateau, and repeat in dyadic blocks with Thue–Morse signs. This single block theorem explains the sign, reflection, complement, zero-window, and peak-height clauses in the supplied Wolfram Language program.

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1 Introduction and scope

The repository [6] develops a detailed connection between the signed Thue–Morse sequence, iterated prefix sums, finite product approximants, the Rvachev up-function, and the Fabius function. In the notation used there, the basic discrete objects are the inclusive prefix sums

$$\sigma_0(m) = \varepsilon_m, \quad \sigma_{r+1}(m) = \sum_{j=0}^m \sigma_r(j),$$

and their formal generating series satisfy

$$\sum_{m \geq 0} \sigma_r(m) z^m = \frac{\mathcal{K}(z)}{(1-z)^r}, \quad \mathcal{K}(z) = \sum_{m \geq 0} \varepsilon_m z^m = \prod_{j \geq 0} (1 - z^{2^j}).$$

The infinite product is interpreted coefficientwise in the formal setting: only finitely many factors affect any specified coefficient. The repository also identifies the finite positive product

$$\prod_{j=1}^{r-1} (1 + z + \cdots + z^{2^j-1})$$

with the first dyadic coefficient block of the order- r prefix row. Those facts are our starting point.

The table in the prompt is not indexed directly by prefix order. Its row recurrence is

$$s_{n,k} = \sum_{j=0}^{k-1} (k-j) s_{n-1,j}, \tag{1}$$

which is a convolution by $1, 2, 3, \dots$. The initial explicit formulas show that values on the lines $k = n + d$ are polynomial in n , but the reason, the general polynomial, and the arithmetic of its coefficients are not apparent from the code alone.

This article has four main goals.

1. Identify the entire two-dimensional table with a shifted subsequence of the iterated prefix-sum array.
2. Derive a closed polynomial formula for every diagonal and explain its basis, degree, leading terms, denominators, and root factors.
3. Prove a complete dyadic block description of every row and use it to explain all acceleration clauses in the supplied program.
4. Give compact Wolfram Language routines for direct evaluation, polynomial generation, root extraction, and independent verification.

The proofs below are self-contained. Statements already present in the repository are cited as inputs; the table identification, diagonal theory, coefficient-content theorem, rational-root restriction, and half-grid root classification are derived here from those inputs. The accompanying Python program performs independent exact checks but is not used as a substitute for proof.

2 Signed Thue–Morse prefixes

2.1 Signs, prefix sums, and conventions

Let $\text{wt}_2(m)$ denote the number of ones in the binary expansion of $m \in \mathbb{N}$. This is the standard signed form of the Thue–Morse sequence [4]; define

$$\varepsilon_m = (-1)^{\text{wt}_2(m)}. \quad (2)$$

The two elementary binary recurrences are

$$\varepsilon_{2m} = \varepsilon_m, \quad \varepsilon_{2m+1} = -\varepsilon_m. \quad (3)$$

We use inclusive prefix sums.

Definition 2.1 (Iterated inclusive prefixes). For $r, m \in \mathbb{N}$, define

$$\sigma_0(m) = \varepsilon_m, \quad \sigma_{r+1}(m) = \sum_{j=0}^m \sigma_r(j). \quad (4)$$

For convenience in shifted formulas, set $\sigma_r(m) = 0$ whenever $m < 0$.

The inclusive convention gives

$$\sigma_{r+1}(m+1) - \sigma_{r+1}(m) = \sigma_r(m+1). \quad (5)$$

The standard hockey-stick argument yields the explicit binomial convolution.

Proposition 2.2 (Binomial convolution). For $r \geq 1$ and $m \geq 0$,

$$\sigma_r(m) = \sum_{a=0}^m \binom{m-a+r-1}{r-1} \varepsilon_a. \quad (6)$$

Proof. For $r = 1$, every binomial coefficient is 1, so the formula is the ordinary prefix sum. If it holds at order r , then

$$\begin{aligned} \sigma_{r+1}(m) &= \sum_{j=0}^m \sum_{a=0}^j \binom{j-a+r-1}{r-1} \varepsilon_a \\ &= \sum_{a=0}^m \varepsilon_a \sum_{u=0}^{m-a} \binom{u+r-1}{r-1} \\ &= \sum_{a=0}^m \binom{m-a+r}{r} \varepsilon_a, \end{aligned}$$

which is the next order by the hockey-stick identity. $\square$

2.2 Formal generating series

Set

$$\mathcal{K}(z) = \sum_{m \geq 0} \varepsilon_m z^m, \quad \mathcal{S}_r(z) = \sum_{m \geq 0} \sigma_r(m) z^m. \quad (7)$$

From Equation (4), multiplication by $(1 - z)^{-1}$ performs one inclusive prefix sum, hence

$$\mathcal{S}_r(z) = \frac{\mathcal{K}(z)}{(1 - z)^r}. \quad (8)$$

The binary recurrences Equation (3) give the Mahler identity

$$\mathcal{K}(z) = (1 - z)\mathcal{K}(z^2), \quad (9)$$

and iteration gives the coefficientwise product

$$\mathcal{K}(z) = \prod_{j \geq 0} (1 - z^{2^j}). \quad (10)$$

Equations (6) and (8) to (10) are the precise discrete ingredients used throughout the rest of the article.

3 The two-dimensional table

3.1 One row step equals two prefix sums

The weighted kernel in Equation (1) has generating series

$$\sum_{h \geq 1} h z^h = \frac{z}{(1 - z)^2}. \quad (11)$$

Thus one row step multiplies a row series by $z(1 - z)^{-2}$: the factor $(1 - z)^{-2}$ performs two prefix summations and the factor z shifts one column to the right. This observation completely identifies the table.

Theorem 3.1 (Exact table identification). *For all $n, k \in \mathbb{N}$, the table computed by the supplied rules satisfies*

$$\boxed{s_{n,k} = \sigma_{2n+1}(k - n - 1).} \quad (12)$$

Consequently its row generating series is

$$T_n(z) := \sum_{k \geq 0} s_{n,k} z^k = \frac{z^{n+1} \mathcal{K}(z)}{(1 - z)^{2n+1}} = \frac{z^{n+1} \mathcal{K}(z^2)}{(1 - z)^{2n}}. \quad (13)$$

Proof. The row $n = 0$ in the code is zero at even columns and equals ε_a at column $2a + 1$. Indeed, the sign-block rule reduces every odd column $2a + 1$ to the value at column 1, while every even column reduces to column 0. On the other hand,

$$\sigma_1(2a) = \varepsilon_a, \quad \sigma_1(2a + 1) = 0,$$

because adjacent Thue–Morse signs cancel. Therefore $s_{0,k} = \sigma_1(k - 1)$, and

$$T_0(z) = z \mathcal{S}_1(z) = \frac{z \mathcal{K}(z)}{1 - z}.$$

Now assume the formula for row $n - 1$. Multiplying by the kernel series Equation (11) gives

$$T_n(z) = \frac{z}{(1 - z)^2} T_{n-1}(z) = \frac{z^{n+1} \mathcal{K}(z)}{(1 - z)^{2n+1}}.$$

The coefficient of z^k is $\sigma_{2n+1}(k - n - 1)$, proving Equation (12).

For a direct index proof, put $q = k - n - 1$. The recurrence right side is

$$\begin{aligned} \sum_{j=0}^{k-1} (k-j) s_{n-1,j} &= \sum_{i=0}^q (q+1-i) \sigma_{2n-1}(i) \\ &= \sum_{u=0}^q \sum_{i=0}^u \sigma_{2n-1}(i) = \sigma_{2n+1}(q), \end{aligned}$$

which is exactly two inclusive prefix summations. Finally, Equation (9) gives the second expression in Equation (13). $\square$

The first rows illustrate both the initial zero triangle and the rapid formation of a positive pulse.

$n \backslash k$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	0	1	0	-1	0	-1	0	1	0	-1	0	1	0	1	0	-1	0
1	0	0	1	2	2	2	1	0	0	0	-1	-2	-2	-2	-1	0	0
2	0	0	0	1	4	9	16	24	32	40	48	55	60	63	64	64	64
3	0	0	0	0	1	6	20	50	104	190	316	490	719	1008	1360	1776	2256

3.2 Bivariate generating functions

Summing Equation (13) over n gives a compact generating function for the whole table.

Corollary 3.2 (Full table series). *As a formal power series in u , with coefficients that are formal power series in z ,*

$$\sum_{n,k \geq 0} s_{n,k} u^n z^k = \frac{z(1-z)\mathcal{K}(z)}{(1-z)^2 - uz}. \quad (14)$$

After shifting to the diagonal coordinate $q = k - n - 1$,

$$\sum_{n,q \geq 0} s_{n,n+q+1} u^n z^q = \frac{(1-z)^2 \mathcal{K}(z^2)}{(1-z)^2 - u}. \quad (15)$$

Proof. Both formulas are geometric sums. For example,

$$\sum_{n \geq 0} u^n T_n(z) = \frac{z\mathcal{K}(z)}{1-z} \sum_{n \geq 0} \left(\frac{uz}{(1-z)^2} \right)^n,$$

which simplifies to Equation (14). Removing the shift z^{n+1} first gives Equation (15). $\square$

4 Exact dyadic block geometry of every row

4.1 The positive pulse polynomial

For an order $r \geq 1$, define

$$\mathcal{A}_r(z) = \prod_{j=1}^{r-1} (1 + z + \cdots + z^{2^j-1}) = \sum_{b=0}^{2^r-r-1} a_{r,b} z^b. \quad (16)$$

The product is empty when $r = 1$, so $\mathcal{A}_1 = 1$. Splitting the first r factors from $\mathcal{K}$ gives an exact block factorization.

Theorem 4.1 (Signed block factorization). *For every $r \geq 1$,*

$$\frac{\mathcal{K}(z)}{(1-z)^r} = \mathcal{A}_r(z) \mathcal{K}(z^{2^r}). \quad (17)$$

Consequently, if $q = a2^r + b$ with $a \geq 0$ and $0 \leq b < 2^r$, then

$$\boxed{\sigma_r(q) = \varepsilon_a a_{r,b}}, \quad (18)$$

where $a_{r,b} = 0$ for $b > 2^r - r - 1$.

Proof. From the iterated Mahler identity,

$$\mathcal{K}(z) = \left(\prod_{j=0}^{r-1} (1 - z^{2^j}) \right) \mathcal{K}(z^{2^r}).$$

The factor $j = 0$ is $1 - z$; for each $1 \leq j < r$, divide $1 - z^{2^j}$ by $1 - z$. This yields Equation (17). Since $\deg \mathcal{A}_r = 2^r - r - 1 < 2^r$, the coefficient of z^{a2^r+b} comes from the unique product of $\varepsilon_a z^{a2^r}$ in $\mathcal{K}(z^{2^r})$ and $a_{r,b} z^b$ in $\mathcal{A}_r$. $\square$

This sharpens the usual terminal-zero statement: it identifies every zero and every nonzero value in every block.

Corollary 4.2 (Complete zero and sign criterion). *For $q = a2^r + b$, $0 \leq b < 2^r$,*

$$\sigma_r(q) = 0 \iff b \geq 2^r - r, \quad (19)$$

$$\operatorname{sgn} \sigma_r(q) = \varepsilon_a \quad \text{whenever } \sigma_r(q) \neq 0. \quad (20)$$

Thus every dyadic block ends in exactly r zeros, and each nonzero block is a positive pulse multiplied by one Thue–Morse sign.

Proof. Every coefficient of $\mathcal{A}_r$ between degrees 0 and $2^r - r - 1$ is strictly positive. One way to see this is the tuple interpretation in Theorem 4.3 below. Outside that range the coefficient is zero. Apply Equation (18). $\square$

4.2 Shape, mass, plateau, and moments

Proposition 4.3 (Complete shape of the positive pulse). *The coefficients of $\mathcal{A}_r$ have the following properties.*

(i) $a_{r,b}$ counts tuples

$$(u_1, \dots, u_{r-1}), \quad 0 \leq u_j \leq 2^j - 1, \quad \sum_{j=1}^{r-1} u_j = b.$$

Hence every coefficient on the full support $0 \leq b \leq 2^r - r - 1$ is a positive integer.

(ii) *The sequence is palindromic:*

$$a_{r,b} = a_{r,2^r-r-1-b}. \quad (21)$$

(iii) *Its total mass is*

$$\sum_b a_{r,b} = \mathcal{A}_r(1) = 2^{\binom{r}{2}}. \quad (22)$$

(iv) It is nondecreasing to the center, constant on exactly r consecutive central indices, and nonincreasing thereafter. The maximum is

$$\max_b a_{r,b} = 2^{\binom{r-1}{2}}, \quad (23)$$

and it occurs for

$$2^{r-1} - r \leq b \leq 2^{r-1} - 1. \quad (24)$$

(v) After normalization by $2^{\binom{r}{2}}$, the coefficients form the probability mass function of

$$U_1 + \cdots + U_{r-1}, \quad U_j \sim \text{Unif}\{0, 1, \dots, 2^j - 1\},$$

with independent summands. Therefore

$$\mathbb{E} \sum U_j = \frac{2^r - r - 1}{2}, \quad (25)$$

$$\text{Var} \sum U_j = \frac{4^r - 3r - 1}{36}. \quad (26)$$

Proof. The tuple interpretation is obtained by choosing one monomial from each factor in Equation (16). The sum of the upper bounds is $\sum_{j=1}^{r-1} (2^j - 1) = 2^r - r - 1$, and convolution of integer intervals has no gaps. Each factor is palindromic, proving (ii). Evaluating the factors at $z = 1$ gives $\prod_{j=1}^{r-1} 2^j = 2^{\binom{r}{2}}$, proving (iii).

For (iv), put $H = 2^{r-1}$. Then

$$\mathcal{A}_r(z) = \mathcal{A}_{r-1}(z)(1 + z + \cdots + z^{H-1}).$$

The degree of $\mathcal{A}_{r-1}$ is $H - r$. For $b < H$,

$$a_{r,b} = \sum_{u=0}^{\min(b, H-r)} a_{r-1,u}. \quad (27)$$

This sum increases strictly until $b = H - r$, then equals the full mass $\mathcal{A}_{r-1}(1) = 2^{\binom{r-1}{2}}$ through $b = H - 1$. Palindromy supplies the descending half. The plateau contains $H - 1 - (H - r) + 1 = r$ indices.

For (v), divide every factor by its value 2^j at 1. The mean of U_j is $(2^j - 1)/2$, and its variance is $((2^j)^2 - 1)/12$. Summing and using geometric-series formulas gives Equations (25) and (26). $\square$

4.3 Specialization to the supplied table

Put

$$r = 2n + 1, \quad B = 2^r = 2^{2n+1}, \quad H = 2^{r-1} = 2^{2n}, \quad c = n + 1. \quad (28)$$

By Equation (12), the shifted coordinate is $q = k - c$. Combining this shift with Equation (18) gives a complete row formula directly in the code's column coordinate.

Theorem 4.4 (Exact first-block pulse and signed repetition). *Write $k = aB + b$, where $0 \leq b < B$. Then*

$$s_{n,k} = \begin{cases} 0, & 0 \leq b \leq n, \\ \varepsilon_a a_{2n+1, b-n-1}, & n+1 \leq b \leq B-n-1, \\ 0, & B-n \leq b < B. \end{cases} \quad (29)$$

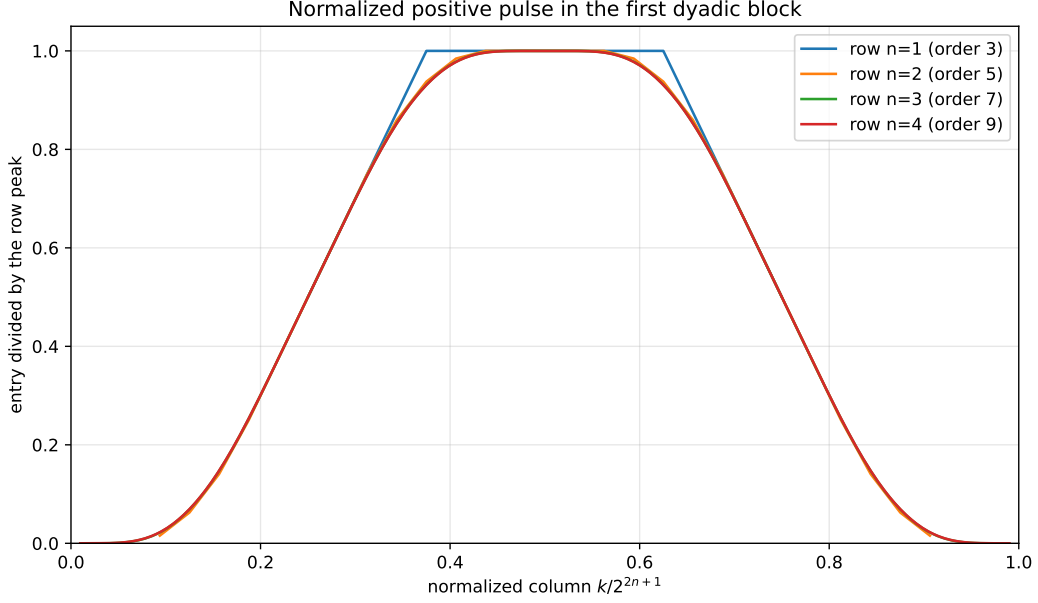


Figure 1: The positive pulse in the first dyadic block of rows $n = 1, 2, 3, 4$, divided by its exact peak $2^{\binom{2n}{2}}$ and plotted against normalized column $k/2^{2n+1}$. The pulse is centered at $1/2$; the exact central plateau has $2n + 1$ lattice points.

Inside every block the nonzero pulse is palindromic about $k = aB + H$, has height

$$M_n = 2^{\binom{2n}{2}}, \quad (30)$$

and has a flat top at the $2n + 1$ columns

$$aB + H - n, \dots, aB + H + n. \quad (31)$$

Adjacent blocks are separated by a zero run of length $2n + 1$, and the sign of block a is ε_a .

Proof. If $b \geq c$, then $q = aB + (b - c)$, so Equation (18) applies with the same block index a . If $b < c$, the shifted residue lies in the terminal zero window of the preceding q -block; both sides of Equation (29) are zero. The end of the support follows from $b - c \leq B - (2n + 1) - 1$, which is equivalent to $b \leq B - n - 1$. The remaining claims are translations of Equations (23) and (24). $\square$

The code's three non-polynomial acceleration rules now become transparent.

Corollary 4.5 (The sign, reflection, and complement rules). *With B, H, M_n as in Equations (28) and (30), the following identities hold whenever the displayed arguments are nonnegative:*

$$s_{n,aB+b} = \varepsilon_a s_{n,b} \quad \text{on the nonzero part of a block,} \quad (32)$$

$$s_{n,k} = s_{n,B-k}, \quad (33)$$

$$s_{n,k} = M_n - s_{n,H-k} \quad (0 \leq k \leq H). \quad (34)$$

The guards in the prompt choose subranges of Equations (33) and (34) on which the new argument is strictly smaller, thereby ensuring rapid recursive reduction.

Proof. The sign rule is Equation (29). For reflection, put $D = \deg \mathcal{A}_{2n+1} = B - 2n - 2$. The pulse coordinate at column k is $k - c$, while the pulse coordinate at $B - k$ is $B - k - c = D - (k - c)$. Thus Equation (33) is precisely the palindromy Equation (21).

For the complement, let $D_- = H - (2n + 1)$, the degree of $\mathcal{A}_{2n}$. The prefix-sum representation Equation (27) and palindromy of $\mathcal{A}_{2n}$ give

$$a_{2n+1,q} = M_n - a_{2n+1,D_- - 1 - q}.$$

Since $D_- - 1 - (k - c) = H - k - c$, this is exactly Equation (34) after translating back to columns. $\square$

Thus the apparently unrelated constants and cut points in the original code are forced by one finite product: B is its repetition period, H its center, M_n its plateau height, and the two reflections are ordinary coefficient palindromy and cumulative-complement symmetry.

5 The general diagonal polynomial

5.1 Definition and three equivalent formulas

Fix an offset $d \geq 1$, and write

$$q = d - 1. \tag{35}$$

The diagonal in the prompt is $k = n + d = n + q + 1$. By Equation (12), its value is $\sigma_{2n+1}(q)$. For fixed q , the prefix order varies with n , and the binomial convolution makes the polynomial dependence explicit.

Theorem 5.1 (Diagonal polynomial theorem). *For every $q \geq 0$, there is a unique polynomial $D_q(x) \in \mathbb{Q}[x]$ such that*

$$s_{n,n+q+1} = D_q(n) \quad (n \in \mathbb{N}). \tag{36}$$

It has the equivalent formulas

$$D_q(x) = \sum_{a=0}^q \varepsilon_a \binom{2x + q - a}{q - a}, \tag{37}$$

$$= \sum_{j=0}^{\lfloor q/2 \rfloor} \varepsilon_j \binom{2x + q - 2j - 1}{q - 2j}, \tag{38}$$

$$= \sum_{j=0}^{\lfloor q/2 \rfloor} \varepsilon_j \frac{(2x)^{\overline{q-2j}}}{(q - 2j)!}. \tag{39}$$

Its ordinary generating function in the diagonal index is

$$\boxed{\sum_{q \geq 0} D_q(x) z^q = \frac{\mathcal{K}(z^2)}{(1 - z)^{2x}} = \frac{\prod_{j \geq 1} (1 - z^{2^j})}{(1 - z)^{2x}}.} \tag{40}$$

For a fixed coefficient q , the product in Equation (40) may be truncated after the last $2^j \leq q$.

Proof. Insert $r = 2n + 1$ and $m = q$ in the binomial convolution Equation (6):

$$s_{n,n+q+1} = \sigma_{2n+1}(q) = \sum_{a=0}^q \varepsilon_a \binom{q - a + 2n}{2n} = \sum_{a=0}^q \varepsilon_a \binom{2n + q - a}{q - a}.$$

Replacing n by an indeterminate x gives Equation (37); each summand is a polynomial, so existence follows. The term $a = 0$ has degree q and every other term has smaller degree, which also proves uniqueness from infinitely many values at $n = 0, 1, 2, \dots$

Alternatively, use the second row-series form in Equation (13). The coefficient at the shifted column is

$$D_q(n) = [z^q] \frac{\mathcal{K}(z^2)}{(1-z)^{2n}}.$$

The coefficient of z^{2j} in $\mathcal{K}(z^2)$ is ε_j , while

$$[z^m] (1-z)^{-2x} = \binom{2x+m-1}{m} = \frac{(2x)^{\overline{m}}}{m!}.$$

Convolution gives Equations (38) and (39), and collecting all coefficients gives Equation (40). $\square$

Remark 5.2 (The requested “general shape”). Formula Equation (39) is more informative than a monomial expansion. The degrees that occur in the natural rising-factorial basis are precisely

$$q, q-2, q-4, \dots,$$

and their signs are the initial Thue–Morse signs $\varepsilon_0, \varepsilon_1, \varepsilon_2, \dots$. Thus the d -th diagonal polynomial is not an isolated interpolation accident: it is the coefficient sequence of one fixed lacunary product multiplied by the negative-binomial family $(1-z)^{-2x}$.

5.2 A closed formula for any table element

The diagonal formula also gives a direct evaluator for any element, without recursing through lower rows.

Corollary 5.3 (Direct table formula). *If $k \leq n$, then $s_{n,k} = 0$. If $k > n$ and $q = k - n - 1$, then*

$$s_{n,k} = \sum_{j=0}^{\lfloor q/2 \rfloor} \varepsilon_j \binom{2n+q-2j-1}{q-2j} \quad (41)$$

$$= \sum_{j=0}^{\lfloor (k-n-1)/2 \rfloor} \varepsilon_j \binom{n+k-2j-2}{k-n-1-2j}. \quad (42)$$

The sum has only $\lfloor (k-n-1)/2 \rfloor + 1$ terms.

This is especially effective near the zero boundary $k = n$: its operation count depends on the diagonal offset, not on the magnitude of n . For a very distant column, first use the signed block reduction Equation (29), then evaluate the coefficient inside one pulse.

5.3 Recovery of the eight displayed formulas

Putting $q = 0, 1, \dots, 7$ in Equation (39) and factoring gives

$$\begin{aligned} D_0(x) &= 1, \\ D_1(x) &= 2x, \end{aligned}$$

$$\begin{aligned}
D_2(x) &= (x+1)(2x-1), \\
D_3(x) &= \frac{2}{3}x(x+2)(2x-1), \\
D_4(x) &= \frac{4x^4 + 12x^3 - x^2 - 3x - 6}{6}, \\
D_5(x) &= \frac{x(x-1)(4x^3 + 24x^2 + 39x + 34)}{15}, \\
D_6(x) &= \frac{(x-1)(x+1)(2x-1)(4x^3 + 32x^2 + 75x + 90)}{90}, \\
D_7(x) &= \frac{x(x-1)(x+2)(2x-1)(4x^3 + 40x^2 + 123x + 192)}{315}.
\end{aligned}$$

Replacing x by n reproduces every initial equation in the prompt. The next sections explain both the visible factors and the denominator sequence $1, 1, 1, 3, 6, 15, 90, 315, \dots$

6 Algebraic structure of the diagonal family

6.1 Monomial coefficients and leading terms

Let $\begin{bmatrix} m \\ \ell \end{bmatrix}$ denote the unsigned Stirling number of the first kind, characterized by

$$y^{\overline{m}} = \sum_{\ell=0}^m \begin{bmatrix} m \\ \ell \end{bmatrix} y^\ell. \quad (43)$$

Expanding Equation (39) gives every monomial coefficient explicitly.

Proposition 6.1 (Exact monomial coefficients). *For $0 \leq \ell \leq q$,*

$$[x^\ell]D_q(x) = 2^\ell \sum_{j=0}^{\lfloor (q-\ell)/2 \rfloor} \varepsilon_j \frac{\begin{bmatrix} q-2j \\ \ell \end{bmatrix}}{(q-2j)!}. \quad (44)$$

In particular,

$$\deg D_q = q, \quad (45)$$

$$[x^q]D_q = \frac{2^q}{q!}, \quad (46)$$

$$[x^{q-1}]D_q = \frac{2^{q-2}}{(q-2)!} \quad (q \geq 2), \quad (47)$$

$$[x^{q-2}]D_q = \frac{2^{q-2}(3q^2 - 7q - 22)}{24(q-2)!} \quad (q \geq 2). \quad (48)$$

Proof. Insert Equation (43) into Equation (39) and collect x^ℓ . Only terms with $q-2j \geq \ell$ contribute, giving Equation (44). The leading term comes only from $j=0$. For the next term, use $\begin{bmatrix} q \\ q-1 \end{bmatrix} = \binom{q}{2}$. For $[x^{q-2}]$, the terms $j=0, 1$ contribute; use

$$\begin{bmatrix} q \\ q-2 \end{bmatrix} = \frac{q(q-1)(q-2)(3q-1)}{24}, \quad \varepsilon_1 = -1,$$

and simplify. □

The first three coefficients already determine useful global statistics of the roots.

Corollary 6.2 (First algebraic moments of the roots). *Let $\rho_1, \dots, \rho_q$ be the complex roots of D_q , counted with multiplicity, where $q \geq 2$. Then*

$$\sum_{i=1}^q \rho_i = -\frac{q(q-1)}{4}, \quad (49)$$

$$\sum_{1 \leq i < j \leq q} \rho_i \rho_j = \frac{q(q-1)(3q^2 - 7q - 22)}{96}, \quad (50)$$

$$\sum_{i=1}^q \rho_i^2 = \frac{q(q-1)(2q+11)}{24}. \quad (51)$$

Thus the arithmetic mean of all roots is $-(q-1)/4$.

Proof. Apply Vieta's formulas to [Equations \(46\) to \(48\)](#). The last identity follows from $(\sum_i \rho_i)^2 = \sum_i \rho_i^2 + 2 \sum_{i < j} \rho_i \rho_j$. $\square$

6.2 Special values and elementary factors

Setting $x = 0$ in the diagonal generating function removes the negative-binomial factor:

$$\sum_{q \geq 0} D_q(0) z^q = \mathcal{K}(z^2). \quad (52)$$

Therefore

$$D_q(0) = \begin{cases} \varepsilon_{q/2}, & q \text{ even,} \\ 0, & q \text{ odd.} \end{cases} \quad (53)$$

In particular, every even-numbered diagonal in the prompt, for which $d = q + 1$ is even, contains the factor x . This explains the factors n in the formulas for $d = 2, 4, 6, 8$.

For even q , Vieta also gives the product of all roots:

$$\prod_{i=1}^q \rho_i = \varepsilon_{q/2} \frac{q!}{2^q}. \quad (54)$$

More generally, the diagonal family is integer-valued on every half-integer.

Proposition 6.3 (Half-integer integrality). *For all $q, m \in \mathbb{N}$,*

$$D_q(m/2) \in \mathbb{Z}. \quad (55)$$

Proof. At $x = m/2$, formula [Equation \(38\)](#) becomes

$$D_q(m/2) = \sum_{j=0}^{\lfloor q/2 \rfloor} \varepsilon_j \binom{m+q-2j-1}{q-2j}.$$

For $m = 0$, the only possible nonzero term has lower index zero. For $m \geq 1$, every binomial coefficient is an integer. $\square$

6.3 The exact coefficient denominator

The denominators displayed in the prompt are not arbitrary. They are the exact least common denominators of the monomial coefficients.

For a nonzero integer polynomial, write $\text{cont } P$ for the greatest common divisor of its coefficients, chosen positive. Recall that a polynomial is primitive if its content is one.

Lemma 6.4 (Content of a doubled rising factorial). *For $m \geq 0$,*

$$\text{cont}((2x)^{\overline{m}}) = 2^{\lceil m/2 \rceil}. \quad (56)$$

After division by this power of two, the reduction modulo 2 has degree $\lceil m/2 \rceil$ and leading coefficient one.

Proof. Put $c = \lceil m/2 \rceil$ and $f = \lfloor m/2 \rfloor$. Separating even and odd factors gives

$$(2x)^{\overline{m}} = 2^c \left(\prod_{a=0}^{c-1} (x+a) \right) \left(\prod_{b=0}^{f-1} (2x+2b+1) \right). \quad (57)$$

Every factor after 2^c is primitive, so their product is primitive by Gauss's lemma. Modulo 2, every factor $2x+2b+1$ becomes 1, while the first product remains monic of degree c . $\square$

Theorem 6.5 (Exact denominator law). *Let $q \geq 0$, put $c = \lceil q/2 \rceil$, and define*

$$E_q(x) = q! D_q(x). \quad (58)$$

Then

$$\boxed{\text{cont } E_q = 2^c}. \quad (59)$$

Consequently the least positive integer δ_q such that $\delta_q D_q(x) \in \mathbb{Z}[x]$ is

$$\boxed{\delta_q = \frac{q!}{2^{\min(\nu_2(q!), \lceil q/2 \rceil)}}}. \quad (60)$$

For $q \geq 4$, this simplifies to

$$\delta_q = \frac{q!}{2^{\lceil q/2 \rceil}}. \quad (61)$$

Proof. From [Equation \(39\)](#),

$$E_q(x) = \sum_{j=0}^{\lfloor q/2 \rfloor} \varepsilon_j \frac{q!}{(q-2j)!} (2x)^{\overline{q-2j}}. \quad (62)$$

For the term indexed by j , [Theorem 6.4](#) contributes the power

$$2^{\lceil (q-2j)/2 \rceil} = 2^{c-j}.$$

The integer $q!/(q-2j)!$ is a product of $2j$ consecutive integers and therefore contains at least j even factors. Every term of [Equation \(62\)](#), and hence E_q , is divisible coefficientwise by 2^c .

It remains to prove that one coefficient is not divisible by 2^{c+1} . Divide the j -th term by 2^c in the form

$$\varepsilon_j \left(\frac{q!/(q-2j)!}{2^j} \right) \left(\frac{(2x)^{\overline{q-2j}}}{2^{c-j}} \right).$$

Both parenthesized expressions are integer-valued polynomials or integers. By [Theorem 6.4](#), the second parenthesis has, modulo 2, degree $c - j$. Thus the coefficient of x^c modulo 2 comes only from $j = 0$, where it is one. Hence $[x^c](E_q/2^c)$ is odd, so the two-adic valuation of the content is exactly c . Finally the leading coefficient of E_q is 2^q ; therefore no odd prime can divide every coefficient. The exact content is consequently 2^c .

The least common coefficient denominator of $E_q/q!$ is obtained by cancelling the greatest common divisor of $q!$ with the content of E_q , which is exactly [Equation \(60\)](#). Finally, $\nu_2(q!) \geq \lceil q/2 \rceil$ for $q \geq 4$. $\square$

For $q = 0, 1, \dots, 15$, the theorem gives

1, 1, 1, 3, 6, 15, 90, 315, 2520, 11340, 113400, 623700, 7484400, 48648600, 681080400, 5108103000,

exactly matching the reduced denominators of the computed polynomials.

6.4 Every rational root is half-integral

The denominator proof contains more two-adic information than is needed to clear coefficients. It sharply restricts rational roots.

Theorem 6.6 (Rational-root restriction). *For $q \geq 1$, every rational root of D_q belongs to $\frac{1}{2}\mathbb{Z}$. In other words, after reduction to lowest terms its denominator is either 1 or 2.*

Proof. Let

$$H_q(x) = \frac{q!}{2^c} D_q(x), \quad c = \lceil q/2 \rceil, \quad f = \lfloor q/2 \rfloor.$$

By [Theorem 6.5](#), $H_q \in \mathbb{Z}[x]$ is primitive. Its leading coefficient is 2^f , so the ordinary rational-root theorem first shows that a reduced root has denominator 2^t for some $0 \leq t \leq f$.

We rule out $t \geq 2$. Write $H_q(x) = \sum_{i=0}^q h_i x^i$. Formula [Equation \(43\)](#) shows that every coefficient of x^i in $E_q = q! D_q$ is divisible by 2^i . Therefore

$$2^{i-c} \mid h_i \quad (i \geq c), \tag{63}$$

and $h_q = 2^f$.

Suppose $a/2^t$, with a odd and $t \geq 2$, were a root. Multiply $H_q(a/2^t) = 0$ by 2^{tq} . The leading term has two-adic valuation exactly f . If $c \leq i < q$, then [Equation \(63\)](#) gives valuation at least

$$(i - c) + t(q - i) = f + (t - 1)(q - i) > f.$$

If $i < c$, the valuation is at least $t(q - i) \geq t(f + 1) > f$. Thus the leading term is the unique term of smallest two-adic valuation, so the sum cannot vanish. This contradiction proves $t \leq 1$. $\square$

Combining [Theorem 6.6](#) with the exact half-grid analysis in the next section completely classifies all *nonnegative* rational roots.

6.5 Difference, derivative, addition, and Mahler identities

It is useful to scale by factorials:

$$\widehat{D}_q(x) = q! D_q(x). \quad (64)$$

Then Equation (40) becomes the exponential generating function

$$\sum_{q \geq 0} \widehat{D}_q(x) \frac{z^q}{q!} = \mathcal{K}(z^2) \exp(-2x \log(1 - z)). \quad (65)$$

Thus $\widehat{D}_q$ is a Sheffer sequence in the standard umbral-calculus sense [5], with invertible factor $\mathcal{K}(z^2)$ and delta series $-2 \log(1 - z)$. No umbral-calculus machinery is required for the identities below; each follows directly from the generating function.

Proposition 6.7 (Half-step and full-step recurrences). *Set $D_{-1} = 0$. For $q \geq 0$,*

$$D_q(x) - D_q(x - \tfrac{1}{2}) = D_{q-1}(x), \quad (66)$$

$$D_q(x + \tfrac{1}{2}) = \sum_{j=0}^q D_j(x), \quad (67)$$

$$D_q(x + 1) = \sum_{j=0}^q (q - j + 1) D_j(x). \quad (68)$$

Proof. Multiplying the generating function at x by $1 - z$ gives the generating function at $x - 1/2$. Taking the difference gives $z \sum_q D_q(x) z^q$, proving Equation (66). Dividing instead by $1 - z$ or $(1 - z)^2$ proves Equations (67) and (68). $\square$

Equation (68) is the diagonal version of the original row recurrence: increasing $x = n$ by one performs the same twofold prefix operation in the diagonal index.

Proposition 6.8 (Derivative and addition laws). *For $q \geq 1$,*

$$D'_q(x) = 2 \sum_{h=1}^q \frac{D_{q-h}(x)}{h}. \quad (69)$$

For arbitrary x, y ,

$$D_q(x + y) = \sum_{j=0}^q D_j(x) \binom{2y + q - j - 1}{q - j}. \quad (70)$$

Proof. Differentiate Equation (40) with respect to x :

$$\frac{\partial}{\partial x} \sum_q D_q(x) z^q = -2 \log(1 - z) \sum_q D_q(x) z^q = 2 \left(\sum_{h \geq 1} \frac{z^h}{h} \right) \left(\sum_q D_q(x) z^q \right).$$

Coefficient comparison gives Equation (69). For addition, factor

$$\frac{\mathcal{K}(z^2)}{(1 - z)^{2(x+y)}} = \frac{\mathcal{K}(z^2)}{(1 - z)^{2x}} (1 - z)^{-2y}$$

and compare coefficients. $\square$

Finally, applying the Mahler relation one level deeper gives a binary self-similarity for the diagonal generating function itself.

Proposition 6.9 (Diagonal Mahler equation). *If $F_x(z) = \sum_{q \geq 0} D_q(x)z^q$, then*

$$\boxed{F_x(z) = (1 - z)(1 + z)^{2x+1} F_x(z^2)}. \quad (71)$$

Equivalently, with

$$g_m(x) = \binom{2x+1}{m} - \binom{2x+1}{m-1} \quad \left(\binom{2x+1}{-1} = 0 \right),$$

one has

$$D_q(x) = \sum_{\substack{0 \leq m \leq q \\ q-m \text{ even}}} g_m(x) D_{(q-m)/2}(x). \quad (72)$$

Proof. Use $\mathcal{K}(z^2) = (1 - z^2)\mathcal{K}(z^4)$ and write

$$F_x(z^2) = \frac{\mathcal{K}(z^4)}{(1 - z^2)^{2x}}.$$

A direct simplification gives Equation (71). The coefficient of z^m in $(1 - z)(1 + z)^{2x+1}$ is $g_m(x)$; coefficient extraction then gives Equation (72). $\square$

7 Half-grid samples and exact rational-root factors

7.1 Diagonal-prefix transposition

At a nonnegative half-integer, the diagonal generating function becomes an ordinary prefix generating function of a different order. This is the key to the root classification.

Theorem 7.1 (Half-grid transposition). *For every $q, m \in \mathbb{N}$,*

$$\boxed{D_q(m/2) = \sigma_{m+1}(q)}. \quad (73)$$

Thus the polynomial D_q , sampled at $0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, reads the fixed column q vertically through all prefix orders $1, 2, 3, 4, \dots$

Proof. By Equation (40) and Equation (9),

$$\sum_{q \geq 0} D_q(m/2)z^q = \frac{\mathcal{K}(z^2)}{(1 - z)^m} = \frac{\mathcal{K}(z)}{(1 - z)^{m+1}} = \sum_{q \geq 0} \sigma_{m+1}(q)z^q.$$

Compare coefficients. $\square$

This identity is a useful conceptual inversion. The original table fixes an odd prefix order $2n+1$ and varies the column. A diagonal polynomial fixes the shifted column q and, at half-integers, varies the prefix order.

7.2 Complete classification of nonnegative rational roots

Apply the complete block zero criterion Equation (19) to the right side of Equation (73) with order $r = m + 1$.

Theorem 7.2 (Exact nonnegative root criterion). *Let $q, m \in \mathbb{N}$. Then*

$$\boxed{D_q(m/2) = 0 \iff q \bmod 2^{m+1} \geq 2^{m+1} - (m + 1).} \quad (74)$$

Equivalently, the lowest $m + 1$ binary digits of q represent one of the last $m + 1$ residues before the next multiple of 2^{m+1} .

If the value is nonzero and

$$q = a2^{m+1} + b, \quad 0 \leq b < 2^{m+1} - (m + 1),$$

then

$$\operatorname{sgn} D_q(m/2) = \varepsilon_a, \quad |D_q(m/2)| = a_{m+1,b} > 0. \quad (75)$$

Proof. By Theorem 7.1, the value is $\sigma_{m+1}(q)$. Now use Equations (18) to (20) with block length 2^{m+1} . $\square$

Because every rational root is half-integral by Theorem 6.6, this result is stronger than a grid statement.

Corollary 7.3 (All nonnegative rational roots). *Define*

$$Z_q = \left\{ m \in \mathbb{N} : q \bmod 2^{m+1} \geq 2^{m+1} - (m + 1) \right\}. \quad (76)$$

Then the nonnegative rational roots of D_q are exactly

$$\left\{ \frac{m}{2} : m \in Z_q \right\}. \quad (77)$$

In particular,

$$L_q(x) := \prod_{m \in Z_q} (2x - m) \quad (78)$$

divides $D_q(x)$ in $\mathbb{Q}[x]$, and L_q contains every linear factor of D_q having a nonnegative rational zero.

The criterion is especially simple in terms of the distance of the low binary suffix from an all-ones word. If

$$b = q \bmod 2^{m+1}, \quad \delta_m(q) = 2^{m+1} - 1 - b,$$

then $m \in Z_q$ exactly when $\delta_m(q) \leq m$.

Proposition 7.4 (Number of nonnegative rational roots). *For $q \geq 1$,*

$$|Z_q| \leq \lceil \log_2(q + 1) \rceil. \quad (79)$$

The bound is attained at every $q = 2^a - 1$, where

$$Z_{2^a-1} = \{0, 1, \dots, a - 1\} \quad (80)$$

and hence

$$\prod_{m=0}^{a-1} (2x - m) \mid D_{2^a-1}(x). \quad (81)$$

More generally, if $q = 2^a - 1 - \delta$ with $0 \leq \delta \leq a - 1$, then

$$\{\delta, \delta + 1, \dots, a - 1\} \subseteq Z_q. \quad (82)$$

Proof. Put $a = \lceil \log_2(q+1) \rceil$, so $q < 2^a$. For $m \geq a$,

$$2^{m+1} - (m+1) \geq 2^{a+1} - (a+1) > 2^a - 1 \geq q,$$

so the root criterion cannot hold. This proves Equation (79).

If $q = 2^a - 1$ and $m < a$, its residue modulo 2^{m+1} is $2^{m+1} - 1$, which is in the zero window. No $m \geq a$ occurs by the previous paragraph, proving Equation (80).

Finally, if $q = 2^a - 1 - \delta$ and $\delta \leq m \leq a - 1$, then

$$q \bmod 2^{m+1} = 2^{m+1} - 1 - \delta \geq 2^{m+1} - (m+1),$$

which proves Equation (82). $\square$

Table 1: Exact nonnegative rational roots for the first sixteen diagonal indices. The table lists Z_q ; the roots themselves are $m/2$.

d	$q = d - 1$	Z_q	forced factor $L_q(x)$
1	0	$\emptyset$	1
2	1	$\{0\}$	$2x$
3	2	$\{1\}$	$2x - 1$
4	3	$\{0, 1\}$	$2x(2x - 1)$
5	4	$\emptyset$	1
6	5	$\{0, 2\}$	$2x(2x - 2)$
7	6	$\{1, 2\}$	$(2x - 1)(2x - 2)$
8	7	$\{0, 1, 2\}$	$2x(2x - 1)(2x - 2)$
9	8	$\emptyset$	1
10	9	$\{0\}$	$2x$
11	10	$\{1\}$	$2x - 1$
12	11	$\{0, 1\}$	$2x(2x - 1)$
13	12	$\{3\}$	$2x - 3$
14	13	$\{0, 2, 3\}$	$2x(2x - 2)(2x - 3)$
15	14	$\{1, 2, 3\}$	$(2x - 1)(2x - 2)(2x - 3)$
16	15	$\{0, 1, 2, 3\}$	$2x(2x - 1)(2x - 2)(2x - 3)$

The criterion also detects irrational positive roots by sign bracketing. For example,

$$D_4(1/2) = -1, \quad D_4(1) = 1,$$

while $Z_4 = \emptyset$. Thus D_4 has a root in $(1/2, 1)$, and Theorem 6.6 proves that this root is irrational. Numerically it is approximately 0.8440034891960957.

7.3 A complete finite-difference test on the negative half-grid

The rational-root restriction reduces the negative side to values at $-m/2$, where m is a positive integer. Unlike the nonnegative side, these values are controlled not by terminal zero windows of prefix sums but by ordinary finite differences of the original sign sequence.

Extend ε_j by zero to negative indices, and write $\nabla f(q) = f(q) - f(q-1)$ for the backward-difference operator.

Proposition 7.5 (Negative half-grid transposition). *For every $q \geq 0$ and every integer $m \geq 1$,*

$$D_q(-m/2) = \sum_{j=0}^{m-1} (-1)^j \binom{m-1}{j} \varepsilon_{q-j} \quad (83)$$

$$= \nabla^{m-1} \varepsilon_q. \quad (84)$$

Consequently a negative rational number is a root of D_q if and only if it is $-m/2$ for some $m \geq 1$ satisfying the finite binomial cancellation in Equation (83). Together with Theorem 7.2, this is a complete exact test for every rational root of every diagonal polynomial.

Proof. Substitute $x = -m/2$ into Equation (40) and use the first Mahler identity once:

$$\sum_{q \geq 0} D_q(-m/2) z^q = \mathcal{K}(z^2)(1-z)^m = \mathcal{K}(z)(1-z)^{m-1}.$$

The coefficient of z^q is exactly Equation (83), with the zero extension taking care of $j > q$. Multiplication by $1-z$ implements one backward difference, which proves Equation (84). The final statement follows from Theorem 6.6. $\square$

Several immediate facts are worth recording. There is never a root at $-1/2$, because $D_q(-1/2) = \varepsilon_q$. A root at -1 occurs exactly when two consecutive Thue–Morse signs agree. More generally, a root at $-m/2$ is a vanishing signed binomial transform of the length- m suffix ending at q . Formula Equation (83) is an $O(\min(m, q))$ exact test and avoids symbolic factorization.

7.4 A provable family of negative integer roots

The nonnegative rational roots are completely classified. Negative roots have a richer structure. The following theorem gives an infinite exact family; it is deliberately stated as a sufficient condition rather than a complete classification.

Theorem 7.6 (Trailing-ones negative roots). *Suppose $q \equiv -1 \pmod{2^a}$. Let ℓ be odd and satisfy $1 \leq \ell \leq a-1$. Then*

$$D_q(-2^\ell) = 0. \quad (85)$$

In particular, for $q = 2^a - 1$,

$$\prod_{\substack{1 \leq \ell \leq a-1 \\ \ell \text{ odd}}} (x + 2^\ell) \text{ divides } D_{2^a-1}(x). \quad (86)$$

Combining with Equation (81),

$$\left(\prod_{m=0}^{a-1} (2x - m) \right) \left(\prod_{\substack{1 \leq \ell \leq a-1 \\ \ell \text{ odd}}} (x + 2^\ell) \right) \mid D_{2^a-1}(x). \quad (87)$$

Proof. Put $h = 2^\ell$. At $x = -h$, the diagonal generating function and Equation (9) give

$$\sum_{q \geq 0} D_q(-h) z^q = \mathcal{K}(z^2)(1-z)^{2h} = \mathcal{K}(z)(1-z)^{2h-1}.$$

Therefore

$$D_q(-h) = \sum_{j=0}^{2h-1} (-1)^j \binom{2h-1}{j} \varepsilon_{q-j}, \quad (88)$$

where terms with $j > q$ are omitted. The hypothesis $\ell \leq a - 1$ gives $2h - 1 < 2^a$. Write $q = Q2^a + (2^a - 1)$. Subtracting any $0 \leq j < 2^a$ affects only the low a bits, so binary complementation gives

$$\varepsilon_{q-j} = \varepsilon_Q \varepsilon_{2^a-1-j} = \varepsilon_Q (-1)^a \varepsilon_j.$$

The fixed nonzero factor $\varepsilon_Q (-1)^a$ may be ignored. Also,

$$(-1)^j \varepsilon_j = \varepsilon_{\lfloor j/2 \rfloor},$$

by checking the even and odd cases with [Equation \(3\)](#). Pairing $j = 2r$ and $j = 2r + 1$ in [Equation \(88\)](#) reduces the remaining sum to

$$\sum_{r=0}^{h-1} \varepsilon_r \binom{2h}{2r+1}. \quad (89)$$

Since $h = 2^\ell$, the map $r \mapsto h - 1 - r$ complements all ℓ binary digits. Thus $\varepsilon_{h-1-r} = (-1)^\ell \varepsilon_r = -\varepsilon_r$, while

$$\binom{2h}{2(h-1-r)+1} = \binom{2h}{2r+1}.$$

The terms of [Equation \(89\)](#) cancel in pairs. □

The theorem explains, for example, the factors $x + 2$ in $D_3, D_7, D_{15}, \dots$, and the additional factor $x + 8$ in $D_{15}, D_{31}, \dots$. Computation reveals further negative integer factors not covered by [Theorem 7.6](#); their complete binary classification is a natural open problem discussed in [Section 11](#).

8 Exact evaluation algorithms in Wolfram Language

8.1 The compact diagonal routine

The sparse rising-factorial formula is the most direct replacement for the eight hand-written diagonal clauses. The following implementation caches a symbolic template for each offset d .

Listing 1: General diagonal polynomial.

```
ClearAll[tmSign, risingPolynomial, diagonalVariable,
  diagonalTemplate, diagonalPolynomial];

tmSign[j_Integer?NonNegative] := (-1)^ThueMorse[j];

risingPolynomial[y_, 0] := 1;
risingPolynomial[y_, m_Integer?Positive] :=
  Product[y + i, {i, 0, m - 1}];

diagonalTemplate[d_Integer?Positive] :=
  diagonalTemplate[d] = Module[{q = d - 1},
    Expand @ Sum[
      tmSign[j]
      risingPolynomial[2 diagonalVariable, q - 2 j]/
      Factorial[q - 2 j],
      {j, 0, Quotient[q, 2]}
    ]
  ];
```

```
diagonalPolynomial[d_Integer?Positive, var_: x] :=
  diagonalTemplate[d] /. diagonalVariable -> var;
```

For example,

```
Table[Factor[diagonalPolynomial[d, n]], {d, 1, 16}]
```

produces the eight formulas in the prompt and continues to arbitrary fixed offset. The exact table identity is simply

```
s[n, n + d] == diagonalPolynomial[d, n]
```

for $n \geq 0$, $d \geq 1$.

A coefficient-extraction implementation is also possible:

```
diagonalPolynomialBySeriesCoefficient[
  d_Integer?Positive, var_: x] := Module[
  {q = d - 1, jmax, finiteK},
  jmax = If[q < 2, 0, IntegerLength[q, 2] - 1];
  finiteK = Product[1 - z^(2^j), {j, 1, jmax}];
  FullSimplify @ SeriesCoefficient[
    finiteK/(1 - z)^(2 var), {z, 0, q}]
];
```

Only factors with exponent $2^j \leq q$ are retained. This form mirrors the generating-function proof, while the rising-factorial form is usually better for explicit polynomial output.

8.2 Direct evaluation of any element

Formula [Equation \(42\)](#) becomes a short exact evaluator.

Listing 2: Direct exact evaluator for any table element.

```
ClearAll[sClosed];

sClosed[n_Integer?NonNegative, k_Integer?NonNegative] /;
  k <= n := 0;

sClosed[n_Integer?NonNegative, k_Integer?NonNegative] :=
  Module[{q = k - n - 1},
    Sum[
      (-1)^ThueMorse[j]
      Binomial[2 n + q - 2 j - 1, q - 2 j],
      {j, 0, Quotient[q, 2]}
    ]
  ];
```

The evaluator has three practical advantages.

1. It does not descend through rows $n - 1, n - 2, \dots$
2. It uses only $\lfloor (k - n - 1)/2 \rfloor + 1$ terms, half as many as the dense binomial convolution.

3. For a fixed diagonal offset, its number of terms is independent of how large n is.

For code that already uses the prompt’s recursive block rules, one may keep those rules and replace all manually entered diagonals by a single seed rule. For example, with a chosen cutoff,

```
$DiagonalPolynomialCutoff = 64;

s[n_Integer?NonNegative, k_Integer?NonNegative] /;
  n < k <= n + $DiagonalPolynomialCutoff :=
  diagonalPolynomial[k - n, n];
```

This gives sixty-four exact polynomial diagonals without storing sixty-four separate formulas. The cutoff is an engineering parameter, not a mathematical limitation.

8.3 Exact block evaluation

For a distant column, the dyadic block theorem may be preferable. The following transparent implementation constructs the pulse polynomial and extracts its coefficient.

Listing 3: Exact signed block evaluator.

```
ClearAll[prefixPulsePolynomial, prefixPulseCoefficient,
  sigmaBlock, sBlockClosed];

prefixPulsePolynomial[r_Integer?Positive] :=
  prefixPulsePolynomial[r] = Expand @ Product[
    Sum[z^u, {u, 0, 2^j - 1}],
    {j, 1, r - 1}
  ];

prefixPulseCoefficient[r_Integer?Positive, b_Integer] /;
  b < 0 || b > 2^r - r - 1 := 0;

prefixPulseCoefficient[r_Integer?Positive,
  b_Integer?NonNegative] :=
  Coefficient[prefixPulsePolynomial[r], z, b];

sigmaBlock[r_Integer?Positive, q_Integer?NonNegative] :=
  Module[{a, b},
    {a, b} = QuotientRemainder[q, 2^r];
    (-1)^ThueMorse[a] prefixPulseCoefficient[r, b]
  ];

sBlockClosed[n_Integer?NonNegative, k_Integer?NonNegative] /;
  k <= n := 0;

sBlockClosed[n_Integer?NonNegative, k_Integer?NonNegative] :=
  sigmaBlock[2 n + 1, k - n - 1];
```

Constructing the whole pulse is appropriate when many entries of one row are needed. Its coefficients can be generated in linear time in the output length by successive sliding-window convolutions with all-ones vectors of length 2, 4, 8, ...; the accompanying Python code uses exactly that method. For a single entry in a very large row, the prompt’s reflection and complement reductions avoid materializing the exponentially long block.

The best hybrid strategy is therefore:

1. use the general diagonal polynomial when $q = k - n - 1$ is small;
2. use signed block reduction when q is large;
3. retain the reflection and complement identities to move a residue toward an already memoized part of the pulse;
4. generate an entire pulse by sliding windows only when a whole block is actually requested.

8.4 Root and denominator utilities

The exact theorems translate directly into inexpensive integer tests.

Listing 4: Exact rational-root tests and coefficient denominator.

```
ClearAll[halfGridRootIndices, halfGridFactor,
  negativeHalfGridValue, negativeHalfGridRootQ,
  minimalMonomialDenominator];

halfGridRootIndices[d_Integer?Positive] := Module[
  {q = d - 1},
  If[q == 0, {},
    Select[Range[0, q - 1],
      Mod[q, 2^(# + 1)] >= 2^(# + 1) - (# + 1) &]
  ]
];

halfGridFactor[d_Integer?Positive, var_: x] :=
  Times @@ (2 var - # & /@ halfGridRootIndices[d]);

(* The point tested here is x=-m/2, m>=1. *)
negativeHalfGridValue[d_Integer?Positive,
  m_Integer?Positive] := Module[{q = d - 1},
  Sum[(-1)^j Binomial[m - 1, j]
    If[q - j < 0, 0, tmSign[q - j]], {j, 0, m - 1}]
];

negativeHalfGridRootQ[d_Integer?Positive,
  m_Integer?Positive] :=
  PossibleZeroQ[negativeHalfGridValue[d, m]];

minimalMonomialDenominator[d_Integer?Positive] := Module[
  {q = d - 1, c, v2},
  c = Ceiling[q/2];
  v2 = IntegerExponent[Factorial[q], 2];
  Factorial[q]/2^Min[v2, c]
];
```

For instance,

```
Table[
  {d, halfGridRootIndices[d],
    Factor[halfGridFactor[d, x]],
    minimalMonomialDenominator[d]},
```

```
{d, 1, 32}
]
```

returns the exact nonnegative rational factor and least coefficient denominator for each of the first thirty-two diagonals without factoring the full polynomial. Negative candidates are tested with `negativeHalfGridRootQ[d,m]`, which evaluates the polynomial at $-m/2$ by the finite-difference formula rather than by expansion.

8.5 Independent consistency checks

The supplied `thue_morse_diagonals.wl` file also contains a literal memoized prefix-sum implementation and the following checks:

```
verifyPromptDiagonals[]
verifyClosedFormula[5, 50]
verifyRecurrence[5, 50]
```

The first compares the general formula with all eight explicit polynomials in the prompt. The second compares the sparse formula, literal repeated prefix sums, and the signed block formula. The third verifies the weighted row recurrence [Equation \(1\)](#). These are regression tests; the preceding sections provide the proofs.

9 Computational experiments and reproducibility

9.1 Exact Python verification

The archive includes `diagonal_analysis.py`. It uses Python integers for the discrete checks and SymPy rational polynomials for symbolic expansion. Running

```
python diagonal_analysis.py --output-dir generated --max-table-q 20
```

creates:

- `verification_report.txt`,
- `diagonal_polynomials.csv`,
- `half_grid_roots.csv`, and
- the PDF and PNG versions of [Figure 1](#).

The bundled run performed the following exact checks.

1. The sparse formula, literal repeated prefix sums, and signed block formula agree for $0 \leq n \leq 5, 0 \leq k \leq 70$.
2. The user’s weighted recurrence agrees on the same range for $1 \leq n \leq 5$.
3. Polynomial evaluation agrees with the integer sparse sum for $0 \leq q \leq 30$.
4. The nonnegative half-grid zero criterion agrees for $0 \leq q, m \leq 30$.

5. The negative half-grid finite-difference formula agrees with exact polynomial evaluation for $0 \leq q \leq 30$, $1 \leq m \leq 16$.
6. The trailing-ones negative-root theorem agrees for exponents $2 \leq a \leq 9$ in four consecutive dyadic blocks.
7. The exact minimal denominator formula agrees with coefficient inspection for $0 \leq q \leq 24$.
8. Pulse support, positivity, palindromy, mass, maximum, and plateau length agree for prefix orders 1 through 10.

All assertions use exact equality. The numerical root quoted after [Theorem 7.2](#) is the only floating-point datum in the main mathematical discussion.

9.2 Selected continued factorizations

The polynomial law continues well beyond the eight hand-entered cases. The following table records exact rational linear factors found by symbolic factorization. The nonnegative factors are predicted a priori by [Theorem 7.2](#); negative factors illustrate the additional structure still to be classified.

Table 2: Selected exact rational factors of D_q .

d	q	rational linear factor product	residual degree
1	0	1	0
2	1	x	0
3	2	$(x+1)(2x-1)$	0
4	3	$x(x+2)(2x-1)$	0
5	4	1	4
6	5	$x(x-1)$	3
7	6	$(x-1)(x+1)(2x-1)$	3
8	7	$x(x-1)(x+2)(2x-1)$	3
9	8	$x+1$	7
10	9	$x(x+2)$	7
11	10	$(x+1)(x+3)(2x-1)$	7
12	11	$x(x+2)(x+4)(2x-1)$	7
13	12	$(x+5)(2x-3)$	10
14	13	$x(x-1)(x+6)(2x-3)$	9
15	14	$(x-1)(x+1)(x+7)(2x-3)(2x-1)$	9
16	15	$x(x-1)(x+2)(x+8)(2x-3)(2x-1)$	9

The factors in [Table 2](#) were verified by exact polynomial division. No irreducibility claim is made for the residual factors unless a computer algebra system has certified it over $\mathbb{Q}$; the table is intended to display recurring linear structure, not to replace the general theorems.

9.3 Fixed-diagonal asymptotics

Because q is fixed on one diagonal, the leading coefficients give a finite asymptotic expansion as $n \rightarrow \infty$:

$$D_q(n) = \frac{(2n)^q}{q!} \left(1 + \frac{q(q-1)}{4n} + \frac{q(q-1)(3q^2-7q-22)}{96n^2} + O_q(n^{-3}) \right). \quad (90)$$

Since D_q is a polynomial, this “asymptotic expansion” terminates after finitely many powers. Formula Equation (44) supplies every remaining coefficient algorithmically. The dominant term has a simple combinatorial interpretation: for fixed q , sufficiently many prefix summations make the initial sign $\varepsilon_0 = 1$ dominate the highest-degree part, while later Thue–Morse signs affect only lower powers of n .

10 Direct connection with the Fabius and Rvachev functions

The finite pulse occurring in row n is not merely analogous to the approximants in the Fabius–Rvachev theory: it is exactly the even-indexed subsequence of those approximants. This identifies the correct shift and normalization of the table without guesswork.

Let up denote Rvachev’s even, compactly supported function, normalized by $\text{up}(0) = 1$, and let F denote the bounded Fabius function on $[0, 1]$, so that

$$F(x) = \text{up}(x - 1) \quad (0 \leq x \leq 1). \quad (91)$$

These are the classical functions introduced from different viewpoints by Fabius and the Rvachevs; their arithmetic relation with signed Thue–Morse prefix sums is developed in detail in [1, 2, 3, 6].

For the table row n , retain

$$B_n = 2^{2n+1}, \quad H_n = 2^{2n}, \quad M_n = 2^{\binom{2n}{2}}. \quad (92)$$

The polynomial $\mathcal{A}_{2n+1}$ of Equation (16) is precisely the finite step polynomial usually denoted $\mathcal{P}_{2n}$. Its degree is

$$G_n = B_n - 2n - 2. \quad (93)$$

For $0 \leq m \leq G_n$, put $k = n + 1 + m$. The standard centered cell coordinate attached to coefficient m is

$$\xi_{n,m} = \frac{2m - G_n}{B_n} = \frac{2k - B_n}{B_n}. \quad (94)$$

Thus the shift by $n + 1$ in Equation (12) exactly cancels the centering correction in the continuous approximation.

Proposition 10.1 (Exact row–histogram identification). *At every coefficient center $\xi_{n,m}$, the normalized centered histogram approximant $\mathcal{W}_{2n}$ satisfies*

$$\mathcal{W}_{2n}(\xi_{n,m}) = \frac{a_{2n+1,m}}{M_n} = \frac{s_{n,n+1+m}}{M_n}. \quad (95)$$

In particular, the positive part of the first table block, after division by its exact height M_n , is the level- $2n$ Rvachev histogram sampled at its cell centers.

Proof. The coefficient identity $a_{2n+1,m} = \sigma_{2n+1}(m)$ in the first dyadic block follows from Equation (18). The table identity $s_{n,n+1+m} = \sigma_{2n+1}(m)$ is Equation (12). Finally, the histogram normalization at level $N = 2n$ is

$$\frac{2^N}{2^{\binom{N+1}{2}}} = \frac{1}{2^{\binom{N}{2}}} = \frac{1}{M_n},$$

and its center formula reduces to Equation (94). $\square$

The relation can be stated without introducing the intervening histogram. It gives a direct approximation algorithm using only the table in the prompt.

Theorem 10.2 (Table limits to up and F). *For $0 \leq x \leq 1$, define*

$$R_n(x) = \frac{s_{n, n+1+\lfloor B_n x \rfloor}}{M_n}, \quad (96)$$

$$F_n(x) = \frac{s_{n, n+1+\lfloor H_n x \rfloor}}{M_n}. \quad (97)$$

Then, pointwise,

$$R_n(x) \longrightarrow \text{up}(2x - 1), \quad (98)$$

$$F_n(x) \longrightarrow F(x). \quad (99)$$

No assertion of a uniform convergence rate is implicit in these formulas.

Proof. By Equation (12),

$$R_n(x) = \frac{\sigma_{2n+1}(\lfloor 2^{2n+1}x \rfloor)}{2^{\binom{2n}{2}}}.$$

This is exactly the corrected prefix-grid sample from the repository at level $2n$. The proved corrected convergence theorem there gives $R_n(x) \rightarrow \text{up}(2x - 1)$ on $[0, 1]$ [6]. Since $F_n(x) = R_n(x/2)$, relation Equation (91) gives

$$F_n(x) \longrightarrow \text{up}(x - 1) = F(x).$$

□

This theorem clarifies the roles of the two limits studied in the article. A fixed row and varying column exhibit the compact pulse and its signed block repetitions. A fixed diagonal and varying row produce the polynomials $D_q(n)$. The Fabius limit instead lets the shifted column q grow exponentially with the row, at the scale $q \asymp 2^{2n}$. Thus fixed-diagonal asymptotics such as Equation (90) and the continuum limit are complementary scaling regimes of the same exact array.

11 Further deductions and research directions

The closed forms above settle the basic diagonal question, but they also expose several sharper problems.

11.1 Binary classification of negative rational factors

The nonnegative rational factors have the residue-window description Equation (74). On the negative side, Equation (83) gives a complete finite-difference test, but it is not yet a comparably transparent description in the binary digits of q and m . A natural next problem is to characterize the zero set

$$\left\{ (q, m) \in \mathbb{N} \times \mathbb{N}_{>0} : \nabla^{m-1} \varepsilon_q = 0 \right\} \quad (100)$$

in terms of carries, complemented suffixes, or a finite collection of binary recursions. The trailing-ones theorem Theorem 7.6 is one infinite stratum of this set.

Because ε_q is a two-automatic sequence, the bivariate zero mask in Equation (100) is a promising candidate for analysis by regular-sequence and linear-representation methods. Even a classification for fixed m , or for m restricted to powers of two, would sharpen the factor tables substantially.

11.2 The residual root geometry

Remove from D_q every rational linear factor detected by Theorems 7.2 and 7.5. The remaining factor can have irrational real roots as well as complex-conjugate pairs. The following questions are concrete and computationally accessible.

1. How many real roots can the residual factor have, and how does this number depend on the binary expansion of q ?
2. Is there a useful vertical strip or parabolic region containing all roots after centering by the exact mean $-(q-1)/4$?
3. Do the empirical root measures have a limiting distribution after an appropriate scaling in q ?
4. Which diagonals have repeated roots? The generating and difference identities provide exact resultant recurrences with which to test square-freeness.

The exact root moments in Equations (49) and (51) give two nontrivial constraints on any proposed limiting law.

11.3 Joint asymptotics beyond a fixed diagonal

Formula Equation (90) treats q as fixed while $n \rightarrow \infty$. The Fabius scaling in Theorem 10.2 instead has $q \asymp 2^{2n}$. Between them lie several unexplored regimes:

$$q = o(n), \quad q \asymp n, \quad q \asymp n^2, \dots, \quad q \asymp 2^{2n}. \quad (101)$$

The coefficient integral

$$D_q(n) = \frac{1}{2\pi i} \oint \frac{\mathcal{K}(z^2)}{(1-z)^{2n}} \frac{dz}{z^{q+1}} \quad (102)$$

provides a common starting point. For subexponential q , the saddle comes mainly from the negative-binomial factor; as it approaches dyadic scales, zeros and oscillations of the lacunary factor $\mathcal{K}(z^2)$ can no longer be treated as a perturbation. A uniform bridge between these regimes would connect the polynomial theory directly to the sharp Fabius asymptotics.

11.4 Fast coefficient queries inside one pulse

Constructing all coefficients of $\mathcal{A}_r$ costs $O(2^r)$ memory, although the sliding-window recurrence is optimal when the whole pulse is required. For one coefficient $a_{r,b}$, it should be possible to exploit the binary expansion of b , palindromy, plateau structure, and recursive interval convolution to obtain substantially smaller memory use. Useful targets include:

- a divide-and-conquer exact evaluator for one coefficient;

- certified modular evaluation followed by Chinese remaindering;
- asymptotically sharp local-limit estimates with rigorous error terms;
- an adaptive method that switches between the sparse diagonal sum and the dyadic pulse representation according to $q = k - n - 1$.

11.5 Formalization

The repository already formalizes the underlying signed sequence, iterated prefix sums, binomial convolution, generating series, dyadic reflection, zero runs, and corrected convergence. The most direct additions suggested by this article are:

1. the exact table identity [Equation \(12\)](#);
2. the diagonal generating function and sparse rising-factorial formula;
3. the content theorem [Theorem 6.5](#);
4. the half-integral rational-root restriction;
5. the positive and negative half-grid transpositions; and
6. the exact plateau and pulse-moment formulas.

Each statement reduces to finite sums, polynomial identities, or formal power-series coefficient extraction and is therefore well suited to a Lean extension of the existing development.

12 Conclusion

The two-dimensional table in the prompt is governed by one exact identity:

$$s_{n,k} = \sigma_{2n+1}(k - n - 1).$$

It converts the weighted row recurrence into two ordinary prefix sums and a shift. From this identity, the complete structure separates into two complementary directions.

Horizontally, each row is a finite positive pulse with generating polynomial

$$\mathcal{A}_{2n+1}(z) = \prod_{j=1}^{2n} (1 + z + \cdots + z^{2^j-1}),$$

repeated in blocks of length 2^{2n+1} with Thue–Morse signs. The pulse is positive, palindromic, unimodal, has an exact $(2n + 1)$ -point plateau, and has height $2^{\binom{2n}{2}}$. These facts explain every reflection, complement, sign, zero-window, and peak constant in the original Wolfram Language program. Under the corrected dyadic scaling, the same rows converge to the Rvachev and Fabius functions.

Diagonally, the line $k = n + d$, with $q = d - 1$, is the degree- q polynomial

$$D_q(x) = \sum_{j=0}^{\lfloor q/2 \rfloor} (-1)^{\text{wt}_2(j)} \frac{(2x)^{\overline{q-2j}}}{(q-2j)!}.$$

This sparse rising-factorial expansion is the general polynomial shape requested in the problem. It gives a direct exact evaluator, all monomial coefficients, the leading asymptotics, the least coefficient denominator, Sheffer and Mahler recurrences, and a complete rational-root test. The initial eight clauses are therefore best viewed as cached specializations of one coefficient formula rather than as separate laws.

A Formula sheet for implementation

For convenient reference, the principal formulas are collected here.

quantity	exact formula
signed Thue–Morse value	$\varepsilon_j = (-1)^{\text{wt}_2(j)}$
order- r prefix series	$\sum_{q \geq 0} \sigma_r(q) z^q = \mathcal{K}(z)/(1-z)^r$
whole table	$s_{n,k} = \sigma_{2n+1}(k-n-1)$
row series	$T_n(z) = z^{n+1} \mathcal{K}(z)/(1-z)^{2n+1}$
full bivariate series	$\sum s_{n,k} u^n z^k = z(1-z) \mathcal{K}(z)/((1-z)^2 - uz)$
positive pulse	$\mathcal{A}_r(z) = \prod_{j=1}^{r-1} (1+z+\dots+z^{2^j-1})$
block law	$\sigma_r(a2^r + b) = \varepsilon_a[z^b] \mathcal{A}_r(z)$
terminal zero test	$\sigma_r(a2^r + b) = 0 \iff b \geq 2^r - r$
diagonal polynomial	$D_q(x) = \sum_{j \leq q/2} \varepsilon_j (2x)^{q-2j} / (q-2j)!$
diagonal generating function	$\sum_{q \geq 0} D_q(x) z^q = \mathcal{K}(z^2)/(1-z)^{2x}$
direct table evaluator	$s_{n,k} = \sum_{j \leq (k-n-1)/2} \varepsilon_j \binom{n+k-2j-2}{k-n-1-2j}$ for $k > n$
minimal coefficient denominator	$q! / 2^{\min(\nu_2(q!), \lceil q/2 \rceil)}$
nonnegative rational root	$m/2$ is a root iff $q \bmod 2^{m+1} \geq 2^{m+1} - (m+1)$
negative rational root	$-m/2$ is a root iff $\sum_{j=0}^{m-1} (-1)^j \binom{m-1}{j} \varepsilon_{q-j} = 0$
Fabius sample	$s_{n,n+1+\lfloor 2^{2n}x \rfloor} / 2^{\binom{2n}{2}} \rightarrow F(x)$

B Reproducibility and archive contents

The accompanying archive contains the following files.

thue_morse_diagonal_polynomials.tex

The complete LaTeX source of this article.

thue_morse_diagonal_polynomials.pdf

The rendered article.

thue_morse_diagonals.wl

Standalone Wolfram Language definitions for the diagonal polynomial, direct table evaluation, pulse/block evaluation, rational-root factors, denominators, and regression tests.

diagonal_analysis.py

Commented exact Python/SymPy verification and data-generation program.

generated/verification_report.txt

Machine-generated pass/fail report for the independent exact checks.

generated/diagonal_polynomials.csv

Expanded and factored polynomials through the configured cutoff.

generated/half_grid_roots.csv

Exact nonnegative rational-root data through the configured cutoff.

figures/normalized_row_pulses.pdf

Vector source for [Figure 1](#); a PNG copy is also included.

From the archive root, the exact computational tables and figure can be regenerated with

```
python diagonal_analysis.py --output-dir generated --max-table-q 20
```

The article can then be rebuilt with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  thue_morse_diagonal_polynomials.tex
```

References

- [1] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174. DOI: 10.1007/BF00536652.
- [2] V. L. Rvachev and V. A. Rvachev, A certain finite function, *Dopovidi Akademii Nauk Ukrain's'koi RSR, Series A* (1971), 705–707, 764. Ukrainian, with English and Russian summaries.
- [3] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Article A51; arXiv:1702.06487, version 3.
- [4] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [5] S. Roman, *The Umbral Calculus*, Pure and Applied Mathematics 111, Academic Press, 1984.
- [6] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction*, Lean 4 formal development and accompanying documentation, consulted 3 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction>.